

SYNOPSIS

Title:

Library Seat Allocation and Real-Time Occupancy Management System

Course Name & Code:

UCS310 – Database Management Systems

Degree & Year:

B.Tech (2nd Year, 4th semester)

Institute Name:

Thapar Institute of Engineering and Technology

Group Members:

- Ayush Jindal – Roll No: 1024030748
- Tanish Bansal – Roll No: 1024030742
- Aditya Kumar Srivastav – Roll No: 1024030745

Lab Instructor Name:

Dr. Banisha Sharma

Academic Year:

2025–2026

1.Introduction

Libraries often face severe seating management issues during peak academic periods such as mid-semester and end-semester examinations. Students spend considerable time traveling to the library only to find no vacant seats or waste additional time searching for available seating. Manual or semi-automated systems fail to provide real-time seat availability information.

This project proposes a **database-driven Library Seat Allocation System** that uses the existing student ID scanning infrastructure to allocate seats automatically on a **first-come-first-serve basis**. The system maintains real-time information regarding seat availability and allocation using a centralized database.

Using a Database Management System instead of a file-based system ensures **data consistency, integrity, concurrent access handling, and efficient query processing**, making it suitable for real-world deployment.

2.Problem Statement

The current library seating process lacks a centralized mechanism to track real-time seat availability and seat occupancy. Students face the following problems:

- No prior information about seat availability
- Time wastage in searching for vacant seats
- Inefficient manual monitoring of seat usage
- Lack of structured historical data for analysis

The scope of database usage in this project includes maintaining student entry/exit records, seat allocation details, and real-time occupancy status to automate the entire seat management process efficiently.

3.Objectives of the Project

The main objectives of this project are:

- To design a database system using **E–R data modeling**
- To convert the ER model into **relational tables**
- To apply **normalization up to Third Normal Form (3NF)**
- To implement the database using **SQL**
- To use **PL/SQL constructs** such as procedures, functions, triggers, and cursors
- To ensure **data consistency and integrity** using constraints and transactions

4.Scope of the Project

The scope of the project is limited to backend database design and implementation.

Types of Users:

- Students (seat allocation via ID scanning)
- Library Administrator (monitoring and management)

Modules Covered:

- Student information management
- Seat information management
- Seat allocation and deallocation
- Real-time availability tracking

5. Proposed System Description

1. Student scans ID at entry
2. System checks seat availability
3. Allocates first available seat (FCFS)
4. Records entry time
5. On exit, seat is released automatically

6. Key features

- Automatic seat allocation on FCFS basis
- Real-time seat availability tracking
- Floor number and seat number assignment
- Centralized and normalized database
- Data integrity through constraints and triggers

7. Database Design

7.1 Entity–Relationship (ER) Diagram

The ER diagram includes the following:

Entities:

- Student
- Floor
- Seat
- Allocation
- Scan_Log

Relationships:

- A student is allocated one seat at a time

- A seat can be allocated to only one student at a time

Constraints:

- One-to-one and one-to-many relationships
- Referential integrity using foreign keys

7.2 Relational Schema

- **STUDENT**(student_id PK, name, course)
- **SEAT**(seat_id PK, floor_no, seat_no, status)
- **ALLOCATION**(allocation_id PK, student_id FK, seat_id FK, entry_time, exit_time)
- **LIBRARY_STATUS**(total_seats, occupied_seats, free_seats)

8.Normalization

Functional dependencies are identified for all tables.

- Tables are converted to **1NF** by ensuring atomic attributes
- **2NF** is achieved by removing partial dependencies
- **3NF** is achieved by eliminating transitive dependencies

The final database design is fully normalized up to **Third Normal Form (3NF)**.

9.Database Implementation

9.1 SQL Implementation

- **DDL Commands:**
CREATE, ALTER, DROP tables
- **DML Commands:**
INSERT, UPDATE, DELETE records
- **SELECT Queries:**
 - Joins to fetch student-seat details
 - Subqueries for seat availability
 - Aggregate functions for occupancy statistics
 - GROUP BY and HAVING clauses

- Views for simplified access

9.2 PL/SQL Components

- Stored procedures for seat allocation and release
- Functions to calculate available seats
- Triggers to update seat status on entry and exit
- Cursors for processing allocation records
- Exception handling for error control

10. Transaction Management & Concurrency

The project uses transaction control commands such as **COMMIT**, **ROLLBACK**, and **SAVEPOINT** to ensure data reliability. ACID properties are maintained to support multiple concurrent users accessing the database without data inconsistency.

11. Tools & Technologies Used

- **Database:** Oracle Database
- **Languages:** SQL, PL/SQL
- **Development Tool:** Oracle SQL Developer
- **Operating System:** Windows

12. Expected Outcomes

- Successful implementation of a normalized database
- Efficient and accurate seat allocation
- Real-time seat availability tracking
- Automation using PL/SQL triggers and procedures
- Improved data consistency, integrity, and reliability

